# Encrypting volumes with NetApp Volume Encryption

## Cloud Manager 3.6

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# Encrypting volumes with NetApp Volume Encryption

NetApp Volume Encryption (NVE) is a software-based technology for encrypting data at rest one volume at a time. Data, Snapshot copies, and metadata are encrypted. Access to the data is given by a unique XTS-AES-256 key, one per volume.

*About this task*

At this time, Cloud Volumes ONTAP supports NetApp Volume Encryption with an external key management server. An Onboard Key Manager is not supported.

You need to set up NetApp Volume Encryption from the ONTAP CLI. You can then use either the CLI or System Manager to enable encryption on specific volumes. Cloud Manager does not support NetApp Volume Encryption from its user interface and from its APIs.

Learn more about supported encryption technologies.

*Steps*

1. Review the list of supported key managers in the NetApp Interoperability Matrix Tool.

   Search for the **Key Managers** solution.

2. Connect to the Cloud Volumes ONTAP CLI.
3. Install a NetApp Volume Encryption license on the Cloud Volumes ONTAP system.

   ONTAP 9 NetApp Encryption Power Guide: Installing the license

4. Install SSL certificates and connect to the external key management servers.

   ONTAP 9 NetApp Encryption Power Guide: Configuring external key management

5. Create a new encrypted volume or convert an existing unencrypted volume using either the CLI or System Manager.
   - CLI:
     - For new volumes, use the **volume create** command with the -encrypt parameter.

       ONTAP 9 NetApp Encryption Power Guide: Enabling encryption on a new volume

     - For existing volumes, use the **volume encryption conversion start** command.

       ONTAP 9 NetApp Encryption Power Guide: Enabling encryption on an existing volume with the volume encryption conversion start command

- System Manager:
  - For new volumes, click **Storage** > **Volumes** > **Create** > **Create FlexVol** and then select **Encrypted**.

    ONTAP 9 Cluster Management using System Manager: Creating FlexVol volumes

  - For existing volumes, select the volume, click **Edit**, and then select **Encrypted**.

    ONTAP 9 Cluster Management using System Manager: Editing volume properties

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